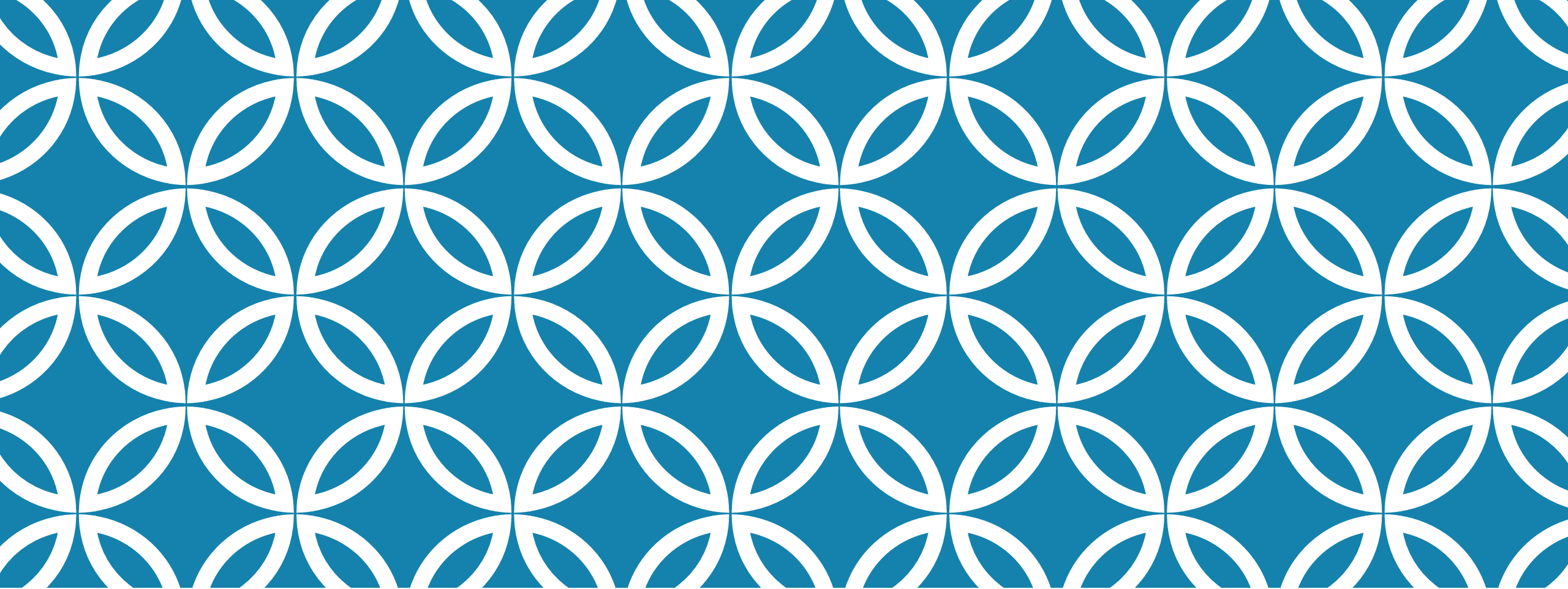
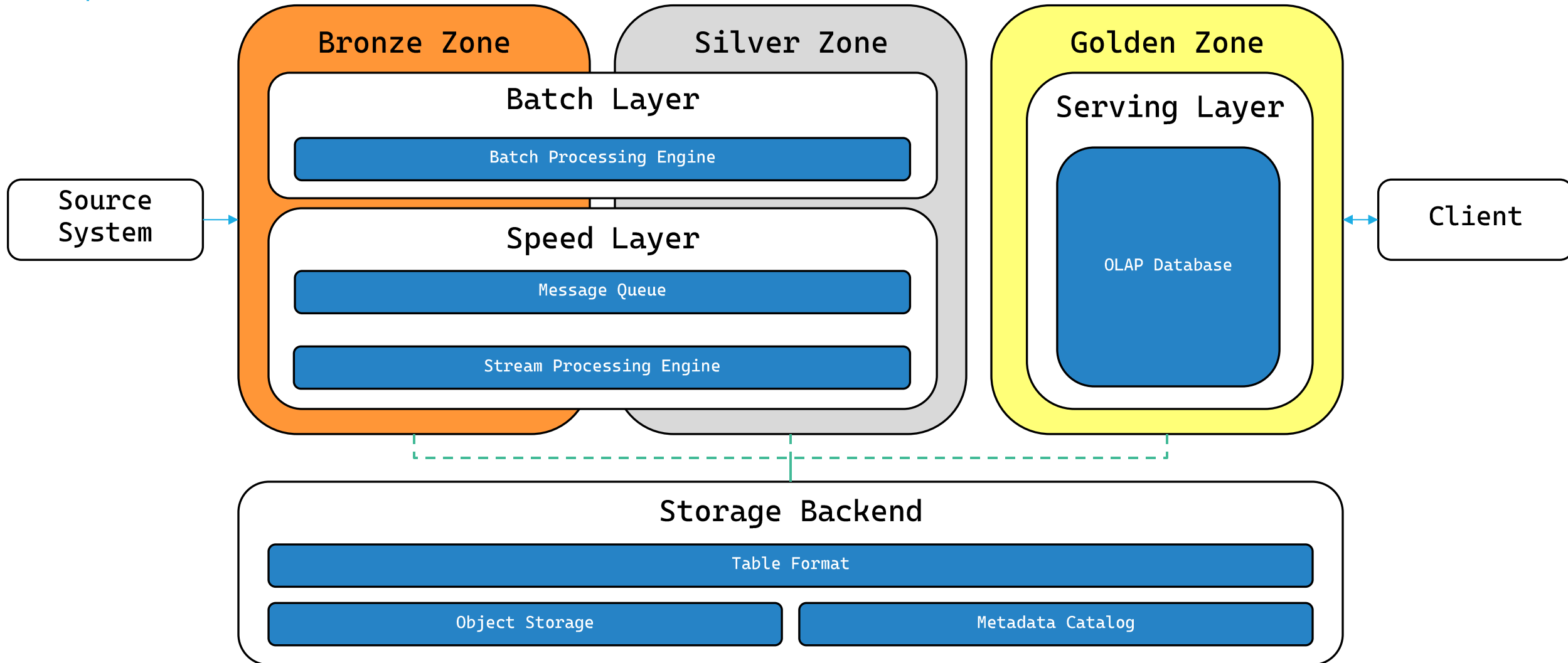




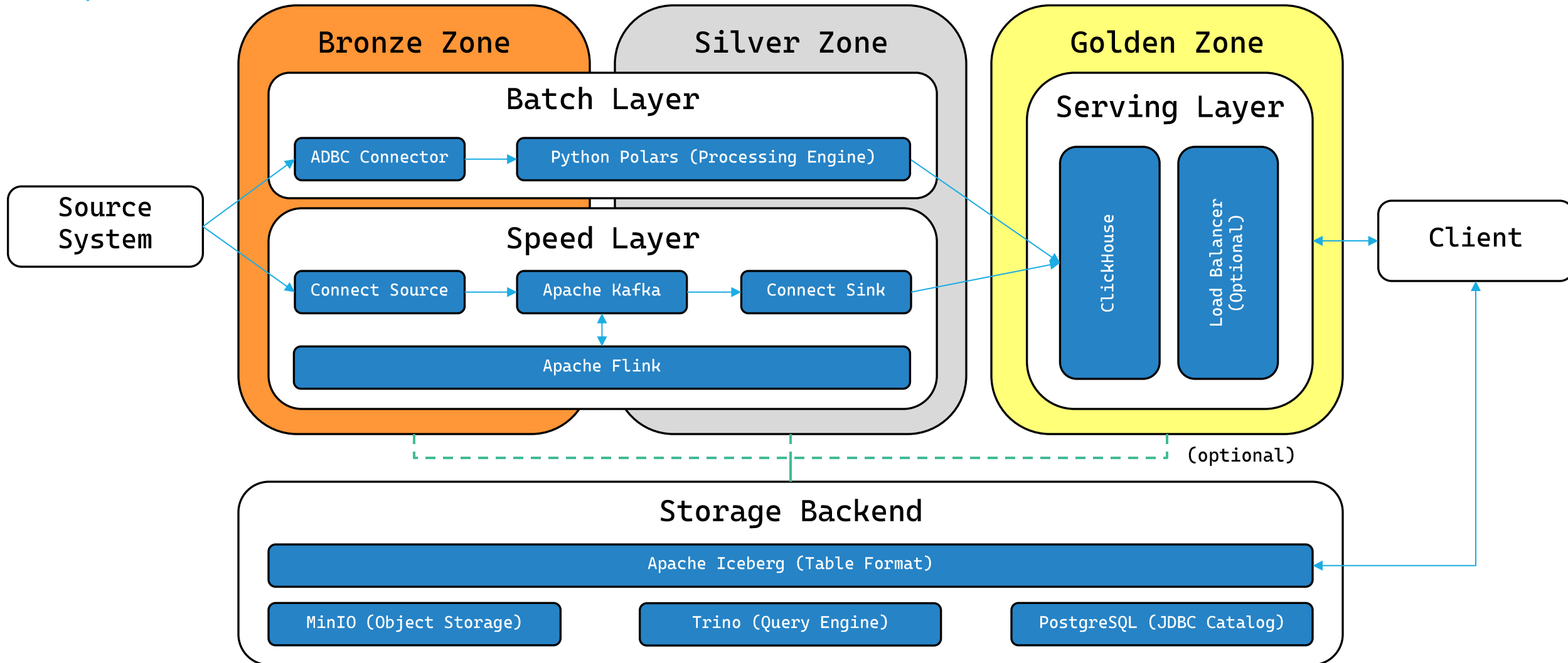
LAMBDA ARCHITECTURE ON TOP OF AN OPEN DATA LAKEHOUSE

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June 2026

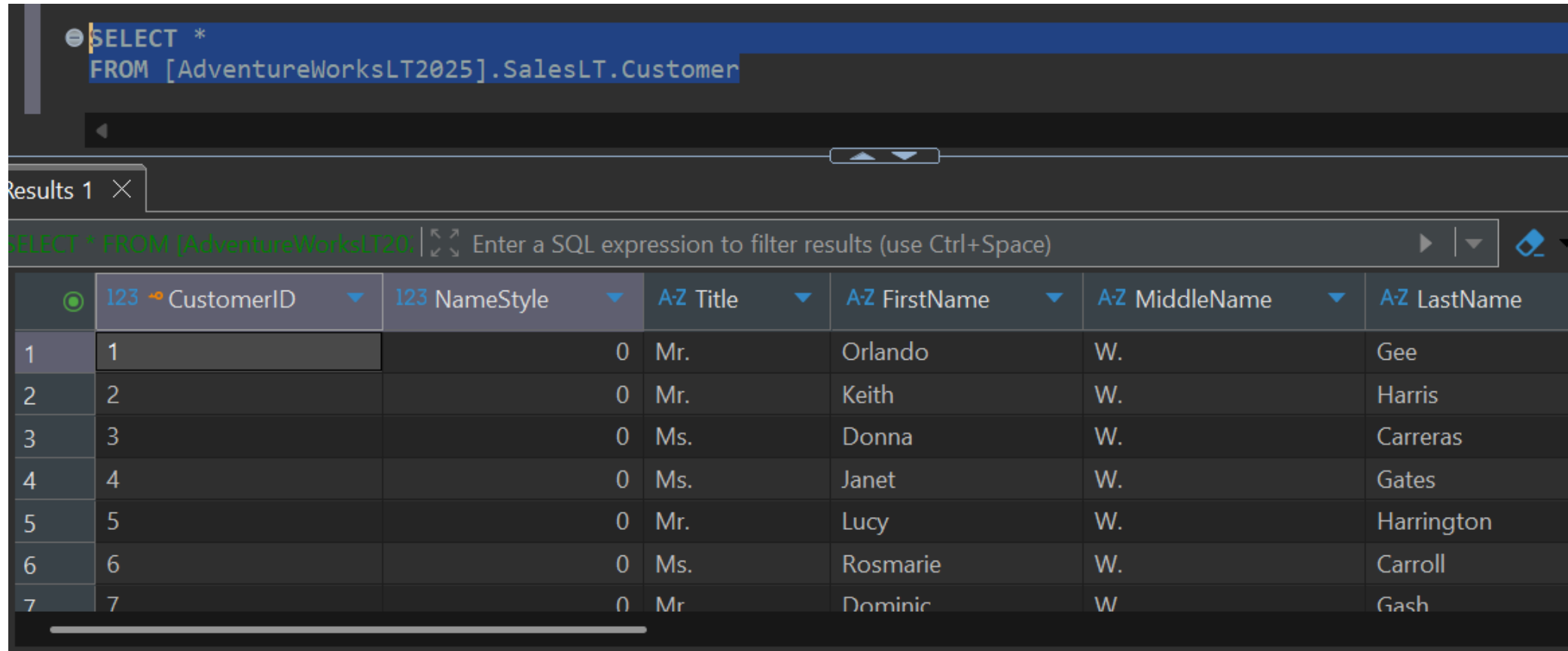
DESIGN OVERVIEW



ACTUAL IMPLEMENTATION



SOURCE DATABASE (SAMPLE)



The screenshot shows a SQL Server Enterprise Manager interface. At the top, a query window displays the following SQL statement:

```
SELECT *  
FROM [AdventureWorksLT2025].SalesLT.Customer
```

Below the query window, the 'Results 1' tab is active, showing a table with 7 rows and 7 columns. The columns are: CustomerID, NameStyle, Title, FirstName, MiddleName, and LastName. The data is as follows:

	123 CustomerID	123 NameStyle	A-Z Title	A-Z FirstName	A-Z MiddleName	A-Z LastName
1	1	0	Mr.	Orlando	W.	Gee
2	2	0	Mr.	Keith	W.	Harris
3	3	0	Ms.	Donna	W.	Carreras
4	4	0	Ms.	Janet	W.	Gates
5	5	0	Mr.	Lucy	W.	Harrington
6	6	0	Ms.	Rosmarie	W.	Carroll
7	7	0	Mr	Dominic	W	Gash

Note:

- Enable CDC for every table used

STAGING TABLE IN BRONZE ZONE (BATCH)

```
$ duckdb
DuckDB v1.5.2 (Variegata)
Enter ".help" for usage hints.
memory D LOAD https;
memory D LOAD iceberg;
memory D CREATE SECRET (
    TYPE s3,
    KEY_ID 'minioadmin',
    SECRET 'minioadmin123',
    REGION 'ap-southeast-1',
    ENDPOINT 'localhost:9100',
    URL_STYLE 'path',
    USE_SSL false
);
```

Success

boolean

true

```
memory D SELECT * FROM iceberg_scan('s3://lakehouse-staging/bronzebatchlayer/customer/metadata/00001-09ff0218-aa68-4e39-a97f-bfcd9a999085.metadata.json') LIMIT 5;
```

buss_date	CustomerID	NameStyle	Title	FirstName	...	Phone	PasswordHash	PasswordSalt	rowguid	ModifiedDate
date	int32	boolean	varchar	varchar	...	varchar	varchar	varchar	varchar	timestamp with time...
2026-06-06	1	false	Mr.	Orlando	...	245-555-0173	L/RLwxzp4w7RWmEgXX+/A7cXa...	1KjXYs4=	3F5AE95E-B87D-4AED-...	2026-06-04 06:58:36...
2026-06-06	2	false	Mr.	Keith	...	170-555-0127	YPdtRdvqeAhj6wyxEsFdshBDN...	fs1ZGhY=	E552F657-A9AF-4A7D-...	2026-06-04 06:58:36...
2026-06-06	3	false	Ms.	Donna	...	279-555-0130	LNoK27abGQo48gGue3EBV/Url...	YTNH5Rw=	130774B1-DB21-4EF3-...	2026-06-04 06:58:36...
2026-06-06	4	false	Ms.	Janet	...	710-555-0173	ELzTpSNbUW1Ut+L5cWlFR7MF6...	nm7D5e4=	FF862851-1DAA-4044-...	2026-06-04 06:58:36...
2026-06-06	5	false	Mr.	Lucy	...	828-555-0186	KJqV15wsX3PG8TS5GSddp6LFF...	cNFKU4w=	83905BDC-6F5E-4F71-...	2026-06-04 06:58:36...

5 rows

memory D

use .last to show entire result

16 columns (10 shown)

Note:

- Scan an Iceberg table is through current metadata file (although you can scan data files directly)
- DELETE/TRUNCATE & INSERT operations
- Bronze zone tables contains 1 date period

STAGING TABLE IN OBJECT STORAGE

The screenshot displays the Minio Object Store Community Edition interface. On the left sidebar, the 'Buckets' section lists four buckets: 'lakehouse', 'lakehouse-rest', 'lakehouse-staging', and 'lakehouse-transform'. The 'lakehouse-staging' bucket is selected and expanded, showing its details: 'Created on: Fri, May 29 2026 21:17:44 (GMT+7)', 'Access: PRIVATE', and '47.4 MiB - 624 Objects'. The main area shows the path 'lakehouse-staging / bronzebatchlayer / customer / data' with a breadcrumb trail. Below the path, a table lists objects with columns for 'Name', 'Last Modified', and 'Size'. The objects are all .parquet files, and the table also shows .json metadata files. The interface includes a search bar at the top right and a 'Create new path' button.

Name	Last Modified	Size
00000-0-0d7e1ca7-0f3c-43aa-933c-1de91a8fa...	Today, 00:24	67.6 KiB
00000-0-2cf285b5-f556-45cd-8dba-ecceff017...	Today, 00:28	61.3 KiB
00000-0-5419c32e-1f2b-494f-8b2f-e21b86b2...	Today, 00:37	62.7 KiB
00000-0-5ca5ef2f-0c5f-43e0-9422-8a86f780...	Today, 00:54	62.6 KiB
00000-0-6f56fe66-9f11-444d-9c83-c244a7ae...	Today, 00:36	62.7 KiB

Note:

- Data file extension is .parquet
- Metadata file extension is .json

TRANSFORMED TABLE IN SILVER ZONE (BATCH)

```
memory D SELECT * FROM iceberg_scan('s3://lakehouse-transform/silverbatchlayer/customer/metadata/00011-2e99bab5-0d24-410c-b94e-7e6cabce4dbd.metadata.json') LIMIT 5;
```

buss_date date	CustomerID int32	NameStyle boolean	Title varchar	FirstName varchar	...	PasswordHash varchar	PasswordSalt varchar	rowguid varchar	ModifiedDate timestamp with time...
2026-06-06	29584	false	Mr.	Walter	...	NN11OHHxrOm6r8lPyhXuBk13QMD+8tph5PIdJ4...	PiILPfc=	BD077ADD-F8F5-471C-...	2026-06-04 06:58:36...
2026-06-06	29585	false	Mr.	Christopher	...	jIw8zSChmDX3xmVInKUV46U77LBgW9C5vIMvK...	J9DNyhU=	11E65E98-1980-4D9F-...	2026-06-04 06:58:36...
2026-06-06	29587	false	Mr.	John	...	7cimopKXrRxK/+QAEW4WJ+XGTEEJvTsX0l1QxC...	C40iUF4=	8916316F-55E7-453E-...	2026-06-04 06:58:36...
2026-06-06	29588	false	Mr.	Willie	...	HKBXG8BXKW53RK2DsPsqNDV7PdDSavcncjD1t/...	dS1y3X0=	DEB25F8C-1D9F-442A-...	2026-06-04 06:58:36...
2026-06-06	29590	false	Mr.	Jo	...	KZaaqy2YCpxZaXM7rzYJyuNkw9w+wWoUaL5P6J...	T2LZ0Po=	7E529A97-2B5C-4339-...	2026-06-04 06:58:36...

5 rows use .last to show entire result 16 columns (9 shown)

```
memory D
```

Note:

- UPSERT (Update & Insert) operations
- Silver zone tables contain historical data
- For convenience, staging and transform table have same structure
- Add any processes as you like (e.g cleansing, enrich, etc)

REAL-TIME PROCESSING (STREAM)

 Apache Flink Dashboard

Overview

Jobs

Running Jobs

Completed Jobs

Task Managers

Job Manager

Submit New Job

Version: 2.2.1 | Commit: 450c63e @ 2026-05-03T10:10:44+02:00 | Message: 0

Flink 2.0 Kafka to ClickHouse

Cancel Job

Job ID	24e0451719857608b08cd1a965f90fa8	Job State	RUNNING 1	Actions	Job Manager Log
Job Type	STREAMING	Start Time	2026-06-07 18:19:29.439	Duration	1h 41m 30s

Overview

Exceptions

Data Skew

TimeLine

Checkpoints

Configuration

Source: Kafka to ClickHouse -> (Sink: Print to Std. Out, Map -> Sink: Writer -> Sink: Committer)

Parallelism: 1

Backpressured (max): 0%

Busy (max): 0%

Data Skew: 0%

Detail

SubTasks

TaskManagers

Watermarks

Accumulators

Back

...

Subtask Metrics

Aggregated Metrics

ID	Bytes Received	Records Received	Byte	Status	More
0	4.92 MB	7,637	4.76 Mi	RUNNING	...

CLICKHOUSE REPLACINGMERGETREE HANDLE RECORD DEDUPLICATION



The screenshot shows the ClickHouse web interface. At the top, the URL is `http://localhost:8223?database=golden`, the version is `v26.6.1.272`, and the uptime is `2 hr`. The default user is `default` and the password is `password`. The command entered in the query editor is `SHOW TABLE saleslt_customer;`. Below the command, a yellow **Run** button is visible, along with the text `(Ctrl/Cmd+Enter, +Shift to run all)` and a checkmark. The execution result shows `1 rows in result, 0.01 sec.` and `Read 1 rows, 844.00 B`. The table structure is displayed in a table with two columns: `Nº` and `statement`. The `statement` column contains the following SQL code:

```
SalesPerson` Nullable(String),
`EmailAddress` Nullable(String),
`Phone` Nullable(String),
`PasswordHash` Nullable(String),
`PasswordSalt` Nullable(String),
`rowguid` Nullable(String),
`ModifiedDate` Nullable(DateTime64(6, 'UTC')),
`__op` Nullable(String),
`__table` Nullable(String),
`__source_ts_ms` Nullable(UInt64)
)
ENGINE = ReplacingMergeTree
PRIMARY KEY (buss_date)
ORDER BY (buss_date, rowguid)
SETTINGS allow_nullable_key = 1, index_granularity = 8192
```

Note:

- `Nullable()` columns is not recommended, this is for testing purposes

RESULTS IN GOLDEN ZONE (BATCH & STREAM USING SAME TABLE DESTINATION)



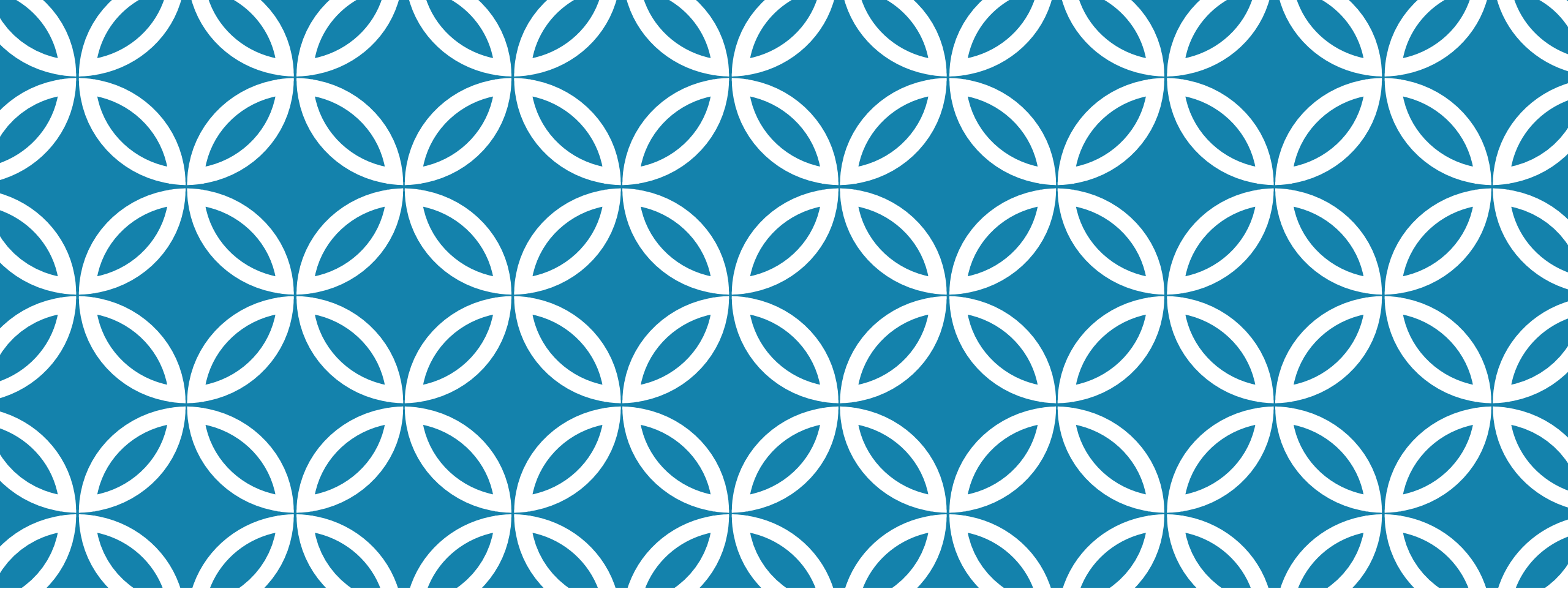
http://localhost:8223?database=goldenv26.6.1.272, uptime 2 hr • defaultpassword

```
SELECT 'saleslt_productmodel', COUNT(1) ROW_COUNT FROM golden.saleslt_productmodel UNION ALL
SELECT 'saleslt_customeraddress', COUNT(1) ROW_COUNT FROM golden.saleslt_customeraddress UNION ALL
SELECT 'saleslt_productmodelproductdescription', COUNT(1) ROW_COUNT FROM golden.saleslt_productmodelproductdescription UNION ALL
SELECT 'saleslt_salesorderheader', COUNT(1) ROW_COUNT FROM golden.saleslt_salesorderheader UNION ALL
SELECT 'saleslt_customer', COUNT(1) ROW_COUNT FROM golden.saleslt_customer UNION ALL
SELECT 'saleslt_address', COUNT(1) ROW_COUNT FROM golden.saleslt_address UNION ALL
SELECT 'saleslt_productdescription', COUNT(1) ROW_COUNT FROM golden.saleslt_productdescription UNION ALL
SELECT 'saleslt_salesorderdetail', COUNT(1) ROW_COUNT FROM golden.saleslt_salesorderdetail UNION ALL
SELECT 'saleslt_productcategory', COUNT(1) ROW_COUNT FROM golden.saleslt_productcategory UNION ALL
SELECT 'saleslt_product', COUNT(1) ROW_COUNT FROM golden.saleslt_product;
```

Run (Ctrl/Cmd+Enter, +Shift to run all) ✓ 10 rows in result, 0.01 sec. Read 10 rows, 160.00 B  

Nº	'saleslt_productmodel'	ROW_COUNT
1	saleslt_address	2250
2	saleslt_productmodel	640
3	saleslt_customeraddress	2085
4	saleslt_productmodelproductdescription	3048
5	saleslt_salesorderheader	160
6	saleslt_customer	2146
7	saleslt_productdescription	3048
8	saleslt_salesorderdetail	2168
9	saleslt_productcategory	205





THANK YOU